

Chapter 9

Sinusoidal Steady-State Analysis

Sinusoids and Phasor

Lec#1	Complex Numbers
Lec#2 9.2	Sinusoids' features
Lec#3 9.3	Phasors
Lec#4 9.4	Phasor relationships for circuit elements
9.5	Impedance and admittance
Lec#5 9.6	Kirchhoff's laws in the frequency domain
9.7	Impedance combinations



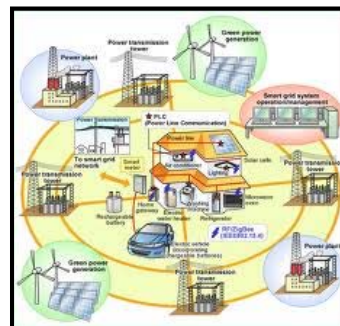
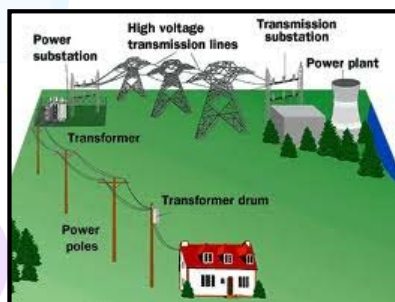
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Motivation

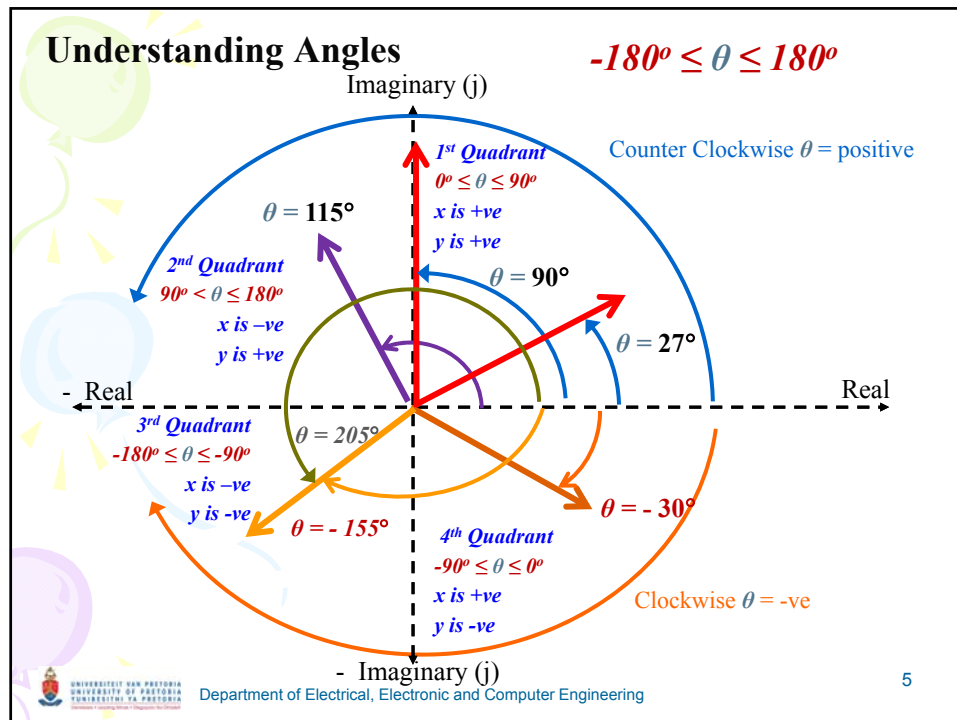
- AC = Alternating Current (Alternates between + and –)
- Vast majority of Power Networks are AC networks because:
 - AC allows for easy stepping up and down of voltages which allows for efficient transfer of energy over long distances.



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Conversion between different forms of Complex Number:

Rectangular: $z = x + j y$

Polar form: $z = r \angle \phi$

Exponential form: $z = r e^{j\phi}$

$r = \sqrt{x^2 + y^2}$

$\phi - \text{phase} = \tan^{-1} \left(\frac{y}{x} \right)$

$z = r \cos \phi + j r \sin \phi$

Convert: $z_1 = 2 + j 6$ into polar and Exponential form

$r = \sqrt{2^2 + 6^2} = 6.32$

$\phi - \text{phase} = \tan^{-1} \left(\frac{6}{2} \right) = 71.57^\circ$

$z_1 = 6.32 \angle 71.57^\circ$ - polar form

$z_1 = 6.32 e^{j71.57}$ - exponential form

Convert: $V = 3.32 \angle 32^\circ$ into rectangular and Exponential form

$x = 3.32 \cos 32^\circ$

$= 2.82$

$y = 3.32 \sin 32^\circ$

$= 1.76$

$V = 2.82 + j 1.76$ - rectangular form

$V = 3.32 e^{j32}$ - exponential form

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Convert: $z_3 = -4.5 + j2$ into polar and Exponential form

$$r = \sqrt{-4.5^2 + 2^2} = 4.92$$

$$\phi - \text{phase} = \tan^{-1}\left(\frac{2}{-4.5}\right) = -23.96^\circ$$

Wrong phase.

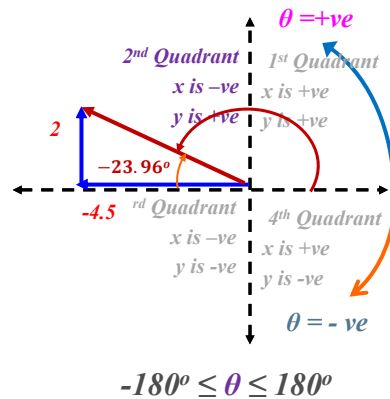
Why ????

x is $-ve$ and y is $+ve$ or 2nd quadrant

$$\therefore \phi = 180 - 23.96^\circ = 156.04^\circ$$

$$z_3 = 4.92 \angle 156.04^\circ - \text{polar form}$$

$$z_3 = 4.92 e^{j156.04} - \text{exponential form}$$



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Convert: $I = 5 \angle 275^\circ$ into rectangular and Exponential form

$$x = 5 \cos 275^\circ$$

Change the angle to $\pm 180^\circ$

$$\theta = 5 \cos(-360^\circ + 275^\circ) = -85^\circ$$

$$x = 5 \cos -85^\circ$$

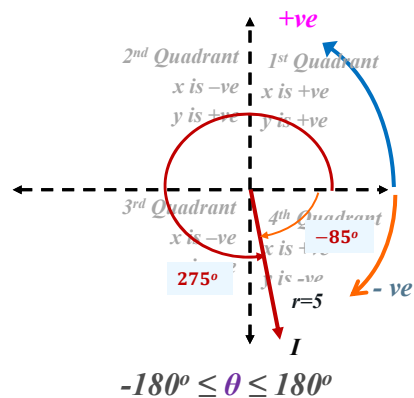
$$= 0.44$$

$$y = 5 \sin -85^\circ$$

$$= -4.98$$

$$I = 0.44 - j4.98 - \text{rectangular form}$$

$$I = 5 e^{-j85^\circ} - \text{exponential form}$$



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Convert: $z_2 = -1 - j 2$ into polar and Exponential form

$$r = \sqrt{-1^2 - 2^2} = 2.24$$

$$\phi - \text{phase} = \tan^{-1} \left(\frac{-2}{-1} \right) = 63.43^\circ$$

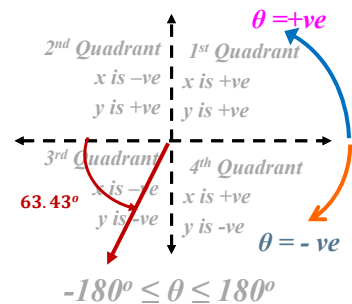
What quadrant is the vector?

x is $-ve$ and y is $-ve$ or 3rd quadrant

$$\therefore \phi = -180 + (63.43^\circ) = -116.57^\circ$$

$$z_2 = 2.24 \angle -116.57^\circ - \text{polar form}$$

$$z_2 = 2.24 e^{j-116.57} - \text{exponential form}$$



Complex number arithmetic

$$\text{Consider: } z_1 = x_1 + jy_1 = r_1 \angle \phi_1^\circ \quad z_2 = x_2 + jy_2 = r_2 \angle \phi_2^\circ$$

$z_1 + z_2 = ?$ **Addition (+):**

MUST WORK IN **RECTANGULAR** FORMAT

$$z = x_1 + jy_1 + x_2 + jy_2$$

$$= (x_1 + x_2) + j(y_1 + y_2)$$

$$\text{Consider: } z_1 = 2.5 + j 4.6 \text{ and } z_2 = 4.5 + j 2$$

Calculate $z_1 + z_2 = ?$

$$z = 2.5 + j 4.6 + 4.5 + j 2$$

$$= (2.5 + 4.5) + j(4.6 + 2)$$

$$= 7 + j 6.6$$

$z_1 - z_2 = ?$ **Subtraction (-):**

MUST WORK IN **RECTANGULAR** FORMAT

$$z = z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

Calculate $z_1 - z_2 = ?$

$$z = (2.5 + j 4.5) - (4.6 + j 2)$$

$$= (2.5 - 4.5) + j(4.6 - 2)$$

$$= -2 + j 2.6$$

Complex number Multiplication (×): **$z_1 z_2 = ?$ WORK IN **POLAR** FORMAT**

Consider: $z_1 = x_1 + jy_1 = r_1 \angle \phi_1^\circ$ $z_2 = x_2 + jy_2 = r_2 \angle \phi_2^\circ$
 $z_1 z_2 = r_1 \angle \phi_1^\circ \cdot r_2 \angle \phi_2^\circ = r_1 \cdot r_2 \angle (\phi_1^\circ + \phi_2^\circ)$

Consider: $z_1 = 2.5 + j4.6$ and $z_2 = 4.5 + j2$. Find $z_1 z_2 = ?$ **Change rectangular to Polar form**

$$z_1 = \sqrt{2.5^2 + 4.6^2} \angle \tan^{-1} \left(\frac{4.6}{2.5} \right)$$

$$= 5.24 \angle 61.48^\circ \text{ - be sure with the quadrant of the angle}$$

$$z_2 = \sqrt{4.5^2 + 2^2} \angle \tan^{-1} \left(\frac{2}{4.5} \right)$$

$$= 4.92 \angle 23.96^\circ \text{ - be sure with the quadrant of the angle}$$

$$r = 5.24 \times 4.92 = 25.78$$

$$\phi = (61.48^\circ + 23.96^\circ) = 85.44^\circ$$

$$z_1 z_2 = 25.78 \angle 85.44^\circ$$

$$\text{Exponential form: } = 25.78 e^{j(85.44^\circ)}$$

Rectangular form:

$$x = 25.78 \cos 85.44^\circ$$

$$y = j 25.78 \sin 85.44^\circ$$

$$z_1 z_2 = 2.05 + j 25.7$$

Method 2:

$$z = (2.5 + j4.6)(4.5 + j2)$$

$$= (2.5 \times 4.5) + (2.5 \times j2)$$

$$+ (j4.6 \times 4.5) + (j4.6 \times j2)$$

$$= 11.25 + j5 + j20.7 + 9.2(j^2)$$

$$= 11.25 + j25.7 + 9.2(-1)$$

$$= 11.25 - 9.2 + j25.7$$

$$= 2.05 + j 25.7$$

Exponential form:

$$r = \sqrt{2.05^2 + 25.7^2} \angle \tan^{-1} \left(\frac{25.7}{2.05} \right)$$

$$= 25.78 e^{j(85.44^\circ)}$$



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Consider: $V = 10 - j5$ and $I = 5 \angle -135^\circ$ **Find $V \cdot I = ?$ Answer in Polar and rectangular form.****Change V to Polar form**

$$V = \sqrt{10^2 + (-5)^2} \angle \tan^{-1} \left(\frac{-5}{10} \right)$$

$$= 11.18 \angle -26.57^\circ$$

- be sure with the quadrant of the angle

$$V \cdot I = 11.18 \times 5 \angle (-26.57^\circ - 135^\circ)$$

$$= 55.9 \angle -161.57^\circ$$

$$= 55.9 \cos -161.57^\circ$$

$$+ j 55.9 \sin -161.57^\circ$$

Rectangular form

$$= -53.1 - j17.7$$

Exponential form

$$= 55.9 e^{j(-161.57^\circ)}$$

Method 2:**Change I to rectangular form**

$$I = 5 \cos -135^\circ + j 5 \sin -135^\circ$$

$$= -3.54 - j3.54$$

$$V \cdot I = (10 - j5)(-3.54 - j3.54)$$

$$= (10 \times -3.54) + (10 \times -j3.54)$$

$$+ (-j5 \times -3.54) + (-j5 \times -j3.54)$$

$$= -35.4 - j35.4 + j17.7 + 17.7(j^2)$$

$$= -35.4 - j17.7 + 17.7(-1)$$

$$= -35.4 - 17.7 - j17.7$$

$$= -53.1 - j 17.7$$

Exponential form:

$$r = \sqrt{-53.1^2 + -17.7^2} \angle \tan^{-1} \left(\frac{-17.7}{-53.1} \right)$$

$$= 55.97 e^{j(-161.53^\circ)}$$



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Complex number Division($\div$): WORK IN **POLAR** FORM

$$\frac{z_1}{z_2} = \frac{r_1 \angle \phi_1}{r_2 \angle \phi_2} = \frac{r_1}{r_2} \angle (\phi_1 - \phi_2)$$

Consider: $z_1 = -2.5 + j 4.6$ and $z_2 = 4.5 - j 2$;

Calculate $\frac{z_1}{z_2} = ?$ Answer in Polar form

Change rectangular to Polar form

$$z_1 = \sqrt{-2.5^2 + 4.6^2} \angle \tan^{-1} \left(\frac{4.6}{-2.5} \right)$$

$$= 5.24 \angle -61.48^\circ$$

$$z_2 = \sqrt{4.5^2 + -2^2} \angle \tan^{-1} \left(\frac{-2}{4.5} \right)$$

$$= 4.92 \angle -23.96^\circ$$

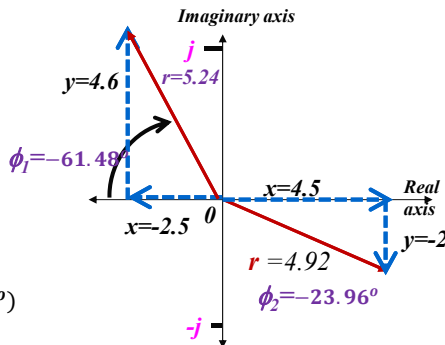
Since z_1 is in 2nd quadrant

$$\phi_1 = 180 - 61.48 = 118.5$$

$$\therefore z_1 = 5.24 \angle 118.5^\circ$$

$$z = \left(\frac{z_1}{z_2} \right) = \frac{5.24}{4.92} \angle (118.5^\circ - (-23.96)^\circ)$$

$$= 1.07 \angle 142.5^\circ$$



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Division: Consider: $z_1 = -2.5 + j 4.6$ and $z_2 = 4.5 - j 2$;

Calculate $\frac{z_1}{z_2} = ?$ Answer in Polar form

$$\frac{z_1}{z_2} = 1.07 \angle 142.5^\circ$$

Method 2:

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{(-2.5 + j4.6)}{(4.5 - j2)} \\ &= \frac{(-2.5 + j4.6)}{(4.5 - j2)} \times \frac{(4.5 + j2)}{(4.5 + j2)} \\ &= \frac{(-2.5 \times 4.5) + (-2.5 \times j2) + (j4.6 \times 4.5) + (j4.6 \times j2)}{(20.25 + 4)} \\ &= \frac{-11.25 - j5 + j20.7 - 9.2}{(20.25 + 4)} \\ &= \frac{-20.45 + j15.7}{24.25} \\ &= -0.84 + j 0.65 \end{aligned}$$

$$\frac{z_1}{z_2} = \sqrt{-0.84^2 + 0.65^2} \angle \tan^{-1} \left(\frac{0.65}{-0.84} \right)$$

$$= 1.06 \angle 142.5^\circ$$



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Complex number **Root** ($\sqrt[n]{Z} = \sqrt[n]{r} \angle \left(\frac{\phi}{n}\right)$): WORK IN **POLAR** FORMAT

Consider: $z_1 = -5 - j 4$

Find $\sqrt{z_1} = ?$

Change z_1 from rectangular to Polar form

$$z_1 = \sqrt{-5^2 + -4^2} \angle \tan^{-1} \left(\frac{-4}{-5}\right) \\ = 6.4 \angle 38.66^\circ$$

- but z_1 is in 3rd quadrant!

$$\phi = -180 + 38.66 = -141.34$$

$$z_1 = 6.4 \angle -141.34^\circ$$

$$\sqrt{z_1} = \sqrt{6.4} \angle \left(\frac{-141.34^\circ}{2}\right)$$

$$\sqrt{z_1} = 2.53 \angle -70.67^\circ$$

$$= 2.53 e^{j(-70.67^\circ)} \text{ - Exponential form}$$

$$= 2.53 \cos -70.67^\circ + j 2.53 \sin -70.67^\circ$$

$$= 0.84 - j 2.39 \text{ - rectangular form}$$



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Consider: $z_1 = -15 + j 4$

Find $\sqrt[3]{z_1} = ?$

Change z_1 from rectangular to Polar form

$$z_1 = \sqrt{-15^2 + 4^2} \angle \tan^{-1} \left(\frac{4}{-15}\right) \\ = 15.52 \angle -14.93^\circ$$

- but z_1 is in 2nd quadrant!

$$\phi = 180^\circ - 14.93^\circ = -165.07^\circ$$

$$z_1 = 15.52 \angle -165.07^\circ$$

$$\sqrt[3]{z_1} = \sqrt[3]{15.52} \angle \left(\frac{-165.07^\circ}{3}\right)$$

$$= 2.5 \angle -55.02^\circ$$

$$= 2.5 e^{j(-55.02^\circ)} \text{ - Exponential form}$$

$$= 2.5 \cos -55.02^\circ + j 2.5 \sin -55.02^\circ$$

$$= 1.433 - j 2.05 \text{ - rectangular form}$$



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Complex number **Inverse** ($\frac{1}{Z} = \frac{1}{r\angle\phi} = \frac{1}{r}\angle -\phi$): WORK IN **POLAR**

FORMAT

Consider: $z_1 = -5 + j4$

Find $\frac{1}{Z_1} = ?$

Change z_1 from rectangular to Polar form

$$z_1 = \sqrt{(-5)^2 + 4^2} \angle \tan^{-1}\left(\frac{4}{-5}\right)$$

$$= 6.4 \angle -38.66^\circ$$

- but z_1 is in 2nd quadrant!

$$\phi = 180 - 38.66 = 141.34$$

$$z_1 = 6.4 \angle 141.34^\circ$$

$$\frac{1}{Z_1} = \frac{1}{6.4 \angle 141.34^\circ}$$

$$= 0.16 \angle -141.34^\circ$$

$$= 0.16 e^{j(-141.34^\circ)} \text{ - Exponential form}$$

$$= 0.16 \cos -141.34^\circ + j 0.16 \sin -141.34^\circ$$

$$= -0.12 - j 0.1 \text{ - rectangular form}$$

Method #2

$$\frac{1}{Z_1} = \frac{1}{-5 + j4} \times \frac{-5 - j4}{-5 - j4}$$

$$= \frac{-5 - j4}{25 - 16j^2}$$

$$= \frac{-5 - j4}{25 - 16(-1)}$$

$$= \frac{-5 - j4}{25 + 16}$$

$$= \frac{-5 - j4}{41}$$

$$= -0.12 - j0.1$$

Complex number **Conjugate** (*):

$Z^* =$ WORK IN **RECTANGULAR** FORMAT

Simply change the sign of “j y”

Consider: $z = -5 + j4$

Find $z^* = ?$

$$z^* = -5 - j4$$

PP 9.3

Evaluate the following complex numbers:

$$x = r \cos \phi \quad y = r \sin \phi$$

$$r = \sqrt{x^2 + y^2} \quad \phi = \tan^{-1} \left(\frac{y}{x} \right)$$

a) $[(5 + j2)(-1 + j4) - 5 \angle 60^\circ]^*$ (Answer in rectangular form)

$$[(5 + j2)(-1 + j4) - 5 \angle 60^\circ]^*$$

$$= [(-5 + j20 - j2 - 8) - 5(\cos 60^\circ + j \sin 60^\circ)]^*$$

$$= [-13 + j18 - 2.5 - j4.33]^*$$

$$= [-15.5 + j13.67]^*$$

To find the conjugate change the sign of the J part.

$$= -15.5 - j13.67$$



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PP 9.3: b) $z = \frac{10 + j5 + 3 \angle 40^\circ}{-3 + j4} + 10 \angle 30^\circ + j5.$

Give answer in all three form

$$= \frac{10 + j5 + 2.3 + j1.93}{5 \angle 126.87^\circ} + 8.66 + j5 + j5$$

$$= \frac{12.3 + j6.93}{5 \angle 126.87^\circ} + 8.66 + j10$$

$$= \frac{14.12 \angle 29.40^\circ}{5 \angle 126.87^\circ} + 8.66 + j10$$

$$= \frac{14.12}{5} \angle (29.4 - 126.87^\circ) + 8.66 + j10$$

$$= 2.82 \angle -97.47^\circ + 8.66 + j10$$

$$= -0.37 - j2.8 + 8.66 + j10$$

$$z = 8.29 + j7.2 \text{ rectangular form}$$

$$\text{Polar form: } z = r \angle \phi$$

$$r = \sqrt{8.29^2 + 7.2^2}$$

$$= 10.98$$

$$\phi = \tan^{-1} \left(\frac{7.2}{8.293} \right)$$

$$= 40.97^\circ$$

$$Z = 10.98 \angle 40.97^\circ$$

$$\text{exponential form:}$$

$$z = 10.98 e^{j40.97^\circ}$$



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Calculate **a** and **b**

$$\sqrt[n]{Z} = \sqrt[n]{r} \angle \left(\frac{\phi}{n} \right)$$

$$2 \angle -35^\circ = \sqrt[2]{\frac{16 \angle 30^\circ - 4 - j8}{a + jb}}$$

$$(2 \angle -35^\circ)^2 = \frac{16 \angle 30^\circ - 4 - j8}{a + jb}$$

$$4 \angle -70^\circ = \frac{13.86 + j8 - 4 - j8}{a + jb}$$

$$a + jb = \frac{9.86}{4 \angle -70^\circ}$$

$$a + jb = \frac{9.86 \angle 0^\circ}{4 \angle -70^\circ}$$

$$a + jb = \frac{9.86}{4} \angle +70^\circ$$

$$a + jb = 2.47 \angle +70^\circ$$

$$= 0.84 + j2.32$$

$$\Rightarrow a = 0.84$$

$$\Rightarrow b = 2.32$$

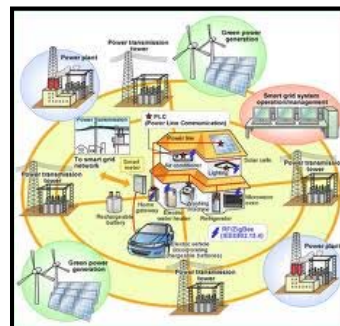
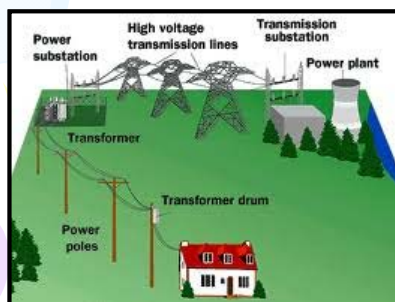


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L#2 Motivation

- AC = Alternating Current (Alternates between + and -)
- Vast majority of Power Networks are AC networks because:
 - AC allows for easy stepping up and down of voltages which allows for efficient transfer of energy over long distances.



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9.2 Sinusoids

- A sinusoid is a signal that has the form of the *sine* or *cosine* function.

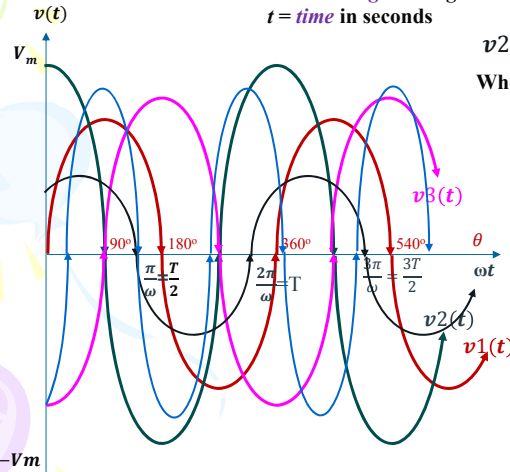
$$v_1(t) = V_m \sin(\theta)$$

where

V_m = the *amplitude* of the sinusoid

θ = the *angle* in degrees

t = *time* in seconds



$$v_2(t) = V_2 \cos(\omega t)$$

Where

ω = the *angular frequency* in radians/s

ωt = *argument* of the sinusoid in radians

T = period [s]

At $t = T$: $\omega T = 2\pi = 360^\circ$

$$T = \frac{2\pi}{\omega}$$

$f = \frac{1}{T}$ *frequency* in [Hz]

$\omega = 2\pi f$ [rad/s]

$$v_3(t) = V_3 \sin(\omega t - \phi)$$

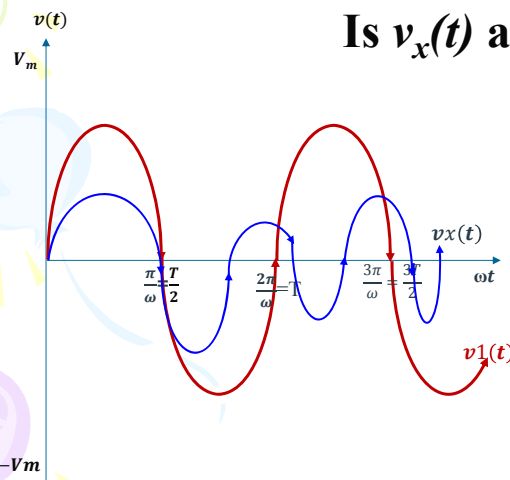
ϕ = the *phase* in radians or degrees

9.2 Sinusoids

- A sinusoid is a signal that has the form of the *sine* or *cosine* function.

$$v_1(t) = V_m \sin(\theta)$$

Is $v_x(t)$ a sinusoid?



PP 9.1

For $f(t) = 5 \sin(4\pi t - 60^\circ)$, determine:

Amplitude; Phase; Angular frequency, Period and Frequency.

Amplitude $= 5$

Phase $\Phi = -60^\circ$
 $= \pi/3 \text{ rad}$

Angular frequency $\omega = 4\pi \text{ rad/s}$

Frequency $\omega = 2\pi f$
 $\therefore f = \frac{\omega}{2\pi}$
 $= \frac{4\pi}{2\pi} = 2 \text{ Hz}$

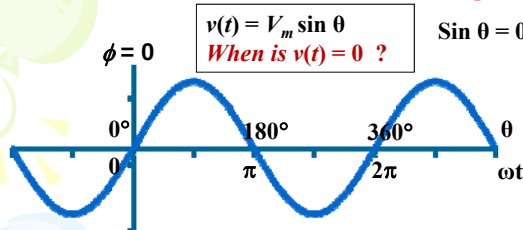
Period $T = \frac{1}{f}$
 $= \frac{2\pi}{\omega} = 0.5 \text{ s}$



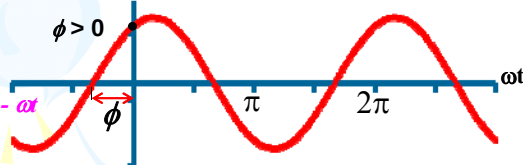
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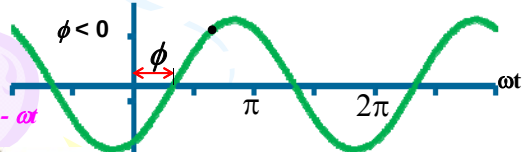
9.2 Sinusoids: Phase Angle ϕ [rad or $^\circ$]



$v(t) = V_m \sin(\omega t + \phi)$
 When is $v(t) = 0$?
 $\sin(\omega t + \phi) = 0$; if $\omega t + \phi = 0$
 But $\phi = 0 \Rightarrow \omega t = 0$
 $\therefore v(t) = 0$ when $t = 0$



When is $v(t) = 0$
 if $\omega t + \phi = 0$
 $\therefore \omega t = -\phi$
 But $\phi > 0 \Rightarrow \omega t = \text{negative}$
 $\therefore v(t) = 0$ when $t < 0$
 • 0-x moves ϕ to left



$v(t) = 0$ if $\omega t + \phi = 0$
 $\therefore \omega t = -\phi$
 But $\phi < 0 \therefore \omega t = \text{positive}$
 $\therefore v(t) = 0$ when $t > 0$
 • 0-x moves ϕ to right



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9.2 Sinusoids

Trigonometric identities: Special Cases

$$\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$$

$$\cos(a \pm b) = \cos a \cos b \mp \sin a \sin b$$

a = ωt & b = $\pi/2 = 90^\circ$

$$\begin{aligned} \cos(\omega t + 90^\circ) &= \cos \omega t \cos 90^\circ - \sin \omega t \sin 90^\circ \\ &= \cos \omega t \times 0 - \sin \omega t \times 1 \\ &= -\sin(\omega t) \Leftrightarrow \cos(\omega t + 90^\circ) \\ \cos(\omega t - 90^\circ) &= \cos \omega t \cos 90^\circ + \sin \omega t \sin 90^\circ \\ &= +\sin(\omega t) \Leftrightarrow \cos(\omega t - 90^\circ) \end{aligned}$$

a = ωt & b = $\pi = 180^\circ$

$$\begin{aligned} -\sin(\omega t) &= \sin(\omega t \pm 180^\circ) \\ \cos(\omega t \pm 180^\circ) &= \cos \omega t \cos 180^\circ - \sin \omega t \sin 180^\circ \\ &= -\cos \omega t \Leftrightarrow \cos(\omega t \pm 180^\circ) \end{aligned}$$

$\sin(\omega t) = -\cos(\omega t - 90^\circ)$

$-\cos(\omega t) = \cos(\omega t \pm 180^\circ)$

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9.2 Sinusoids

Graphical approach: Example

1. Plot $i(t) = \cos(\omega t + 90^\circ)$
2. Plot $i(t) = \cos(\omega t - 90^\circ)$
3. Plot $v(t) = \sin(\omega t + 60^\circ)$
3. Plot $f(t) = \cos(\omega t - 260^\circ)$

$-\sin(\omega t) \Leftrightarrow \cos(\omega t + 90^\circ)$
 $+\sin(\omega t) \Leftrightarrow \cos(\omega t - 90^\circ)$
 $-\cos \omega t \Leftrightarrow \cos(\omega t \pm 180^\circ)$

$\cos(180^\circ) = -1$
 $\cos(\omega t + 180^\circ) = -1$
 $-\cos(\omega t)$

$f(t) = -\sin(\omega t + 10^\circ)$
 ~~$f(t) = \cos(\omega t - 260^\circ)$~~
 $f(t) = \cos(\omega t + 100^\circ)$

$\cos(0^\circ) = 1$
 $\cos(\omega t + 0^\circ) = 1$
 $\cos(\omega t)$

Counter Clockwise = positive phase shift

Clockwise = -ve phase shift

$v(t) = \cos(\omega t - 30^\circ)$
 $v(t) = \sin(\omega t + 60^\circ)$

$i(t) = \cos(\omega t - 90^\circ)$
 $\sin(\omega t)$
 $\cos(\omega t - 90^\circ) \Leftrightarrow \sin(\omega t)$

$-180^\circ \leq \phi \leq 180^\circ$

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All Sinusoids MUST be represented using

$$f(t) = A_m \cos(\omega t \pm \phi^\circ)$$

Unless clearly stated

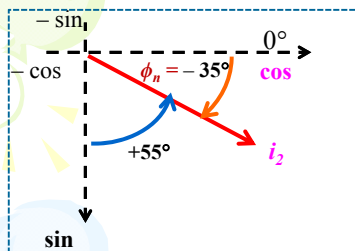


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Transforming sinusoids ($+A_m \sin \Rightarrow A_m \cos$)



Change $i_2 = 5 \sin(\omega t + 55^\circ)$

to $i_2 = 5 \cos(\omega t + \phi_n)$

$$i_2 = 5 \cos(\omega t - 35^\circ)$$

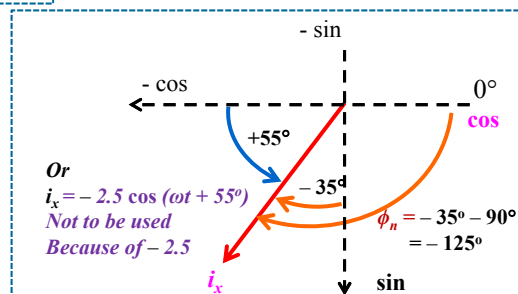
$$+A_m \sin(\omega t \pm \phi) \Rightarrow +A_m \cos(\omega t \pm \phi - 90^\circ)$$

Change $i_x = 2.5 \sin(\omega t - 35^\circ)$

to $i_x = 2.5 \cos(\omega t + \phi_n)$

$$i_x = 2.5 \cos(\omega t - 125^\circ)$$

$$-180^\circ \leq \phi \leq 180^\circ$$



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Transforming sinusoids ($-A_m \sin \Rightarrow +A_m \cos$)

Change $v_2 = -0.5 \sin(\omega t + 55^\circ)$

$v_2 = +0.5 \cos(\omega t + \phi_n)$

$v_2 = 0.5 \cos(\omega t + 125^\circ)$

$-A_m \sin(\omega t \pm \phi) \Rightarrow +A_m \cos(\omega t \pm \phi + 90^\circ)$

Change $v_x = -100 \sin(\omega t - 40^\circ)$

$v_x = +100 \cos(\omega t + \phi_n)$

$v_x = 100 \cos(\omega t + 50^\circ)$

$-180^\circ \leq \phi \leq 180^\circ$

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Transforming sinusoids ($-A_m \cos \Rightarrow +A_m \cos$)

Change $v_2 = -20.5 \cos(\omega t - 130^\circ)$

$$v_2 = +20.5 \cos(\omega t \pm \phi_n)$$

$$v_2 = 20.5 \cos(\omega t + 50^\circ)$$

$-A_m \cos(\omega t \pm \phi) \Rightarrow +A_m \cos(\omega t \pm \phi \pm 180^\circ)$

Change $i_x = -100 \cos(\omega t + 40^\circ)$

$$i_x = +100 \cos(\omega t \pm \phi_n)$$

$$i_x = 100 \cos(\omega t - 140^\circ)$$

$$-180^\circ \leq \phi \leq 180^\circ$$

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Sinusoids: Leading or lagging? Graphical approach

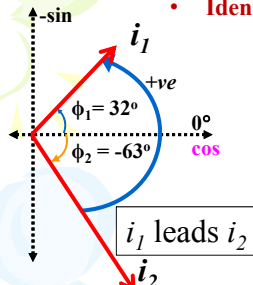
Consider: $i_1 = 8 \cos(\omega t + 32^\circ)$ and $i_2 = 20.5 \cos(\omega t - 63^\circ)$

- Identify which sinusoid is leading?

- Always look at angle from **cos** axis and $< 180^\circ$
- Remember: $\curvearrowright$ = positive
- Vectors rotate counter clockwise,

So ask yourself: "Which one passes 0° first?"

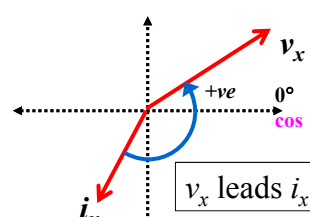
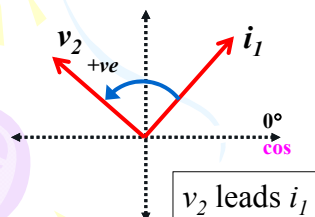
(Both vectors must start in bottom two quadrants!)



Or

$$\Phi_{i1-i2} = \phi_1 - \phi_2 \text{ if } > 0, \text{ then } i_1 \text{ leads } i_2$$

$$\Phi_{i2-i1} = \phi_2 - \phi_1 \text{ if } > 0, \text{ then } i_2 \text{ leads } i_1$$



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PP 9.2 Determine the phase angle between $i_1 = -4\sin(377t + 25^\circ)$ and $i_2 = 5 \cos(377t - 40^\circ)$. Indicate which sinusoid leads.

Analytical:

Change all signals to **+ cos**

$$+A_m \sin(\omega t \pm \phi) \Rightarrow +A_m \cos(\omega t \pm \phi - 90^\circ)$$

$$-A_m \sin(\omega t \pm \phi) \Rightarrow +A_m \cos(\omega t \pm \phi + 90^\circ)$$

$$-A_m \cos(\omega t \pm \phi) \Rightarrow +A_m \cos(\omega t \pm \phi \pm 180^\circ)$$

$$i_1 = -4 \sin(377t + 25^\circ)$$

$$\Rightarrow +4 \cos(377t + 25^\circ + 90^\circ)$$

$$= 4 \cos(377t + 115^\circ)$$

$$\Delta\phi = \phi_1 - \phi_2$$

$$\Delta\phi > 0 \Rightarrow i_1 \text{ leads } i_2$$

$$\text{or } i_2 \text{ lags } i_1$$

$$\Delta\phi < 0 \Rightarrow i_2 \text{ leads } i_1$$

$$\text{or } i_1 \text{ lags } i_2$$

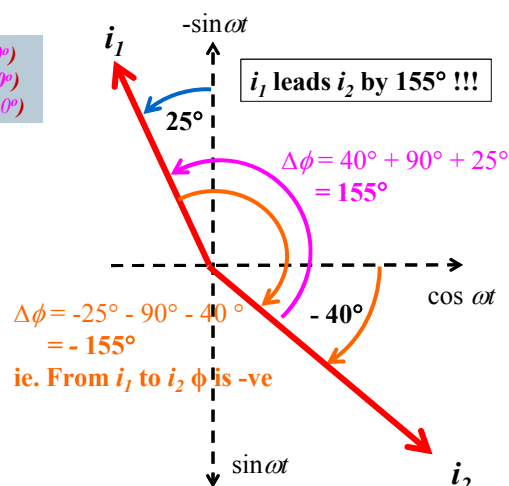
$$\Delta\phi = 115^\circ - (-40^\circ)$$

$$= 155^\circ$$

(Always give angle $< 180^\circ$)

Hence: i_1 leads i_2

Graphical: Plot



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L#3

How does one analyse an AC circuit?

Let: $i_1 = -4 \sin(377t + 25^\circ)$ and $i_2 = 5 \cos(377t - 40^\circ)$. Calculate: i_s



$$\begin{aligned}\sin(a \pm b) &= \sin a \cos b \pm \cos a \sin b \\ \cos(a \pm b) &= \cos a \cos b \mp \sin a \sin b\end{aligned}$$

Apply KCL: $i_s = i_1 + i_2$

$$i_s = -4 \sin(377t + 25^\circ) + 5 \cos(377t - 40^\circ)$$

$$\begin{aligned}i_s &= -4(\sin 377t \cos 25^\circ + \cos 377t \sin 25^\circ) + \\ &\quad + 5(\cos 377t \cos -40^\circ - \sin 377t \sin -40^\circ) \\ &= -4(0.91 \sin 377t + 0.42 \cos 377t) + \\ &\quad + 5(0.77 \cos 377t - -0.64 \sin 377t) \\ &= -3.64 \sin 377t + 1.68 \cos 377t + 3.85 \cos 377t + \\ &\quad 3.2 \sin 377t \\ &= -0.44 \sin 377t + 5.53 \cos 377t\end{aligned}$$



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Current flowing through an element is $i = -4 \sin(377t + 25^\circ)$ and voltage associated with this element is $v = 5 \cos(377t - 40^\circ)$. What is the power associated with the element?

$$\begin{aligned}\sin(a \pm b) &= \sin a \cos b \pm \cos a \sin b \\ \cos(a \pm b) &= \cos a \cos b \mp \sin a \sin b\end{aligned}$$

$$\text{Power} = i \times v$$

$$= -4 \sin(377t + 25^\circ) \times 5 \cos(377t - 40^\circ)$$

$$\begin{aligned}P &= -20(\sin 377t \cos 25^\circ + \cos 377t \sin 25^\circ) \times \\ &\quad (\cos 377t \cos -40^\circ - \sin 377t \sin -40^\circ) \\ &= -20(0.91 \sin 377t + 0.42 \cos 377t) \times \\ &\quad (0.77 \cos 377t - -0.64 \sin 377t) \\ &= -20(1.55 \sin 377t + 1.19 \cos 377t)\end{aligned}$$

Is there another method?

Yes

Phasor

By changing time based sinusoids
to frequency or **phase domain**
Complex number



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How does one analyse an AC circuit?

Method #1:
Time domain

Method #2:
Frequency or
Phasor domain

KVL:

$$-V_s(t) + v_R(t) + v_C(t) + v_L(t) = 0$$

$$-10 \cos 4t + iR + \frac{1}{C} \int_{t_0}^t i dt + L \frac{di}{dt} = 0$$

Eliminate $\int \frac{d}{dt} i(t)$:

$$\frac{d}{dt}(-10 \cos 4t) + \frac{d}{dt}(iR) + \frac{d}{dt}\left(\frac{1}{C} \int_{t_0}^t i dt\right) + \frac{d}{dt}\left(L \frac{di}{dt}\right) = 0$$

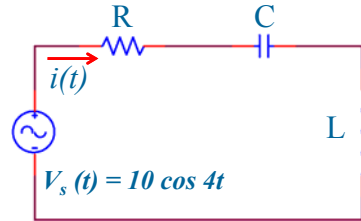
$$40 \sin 4t + R \frac{di}{dt} + \frac{i}{C} + L \frac{d^2 i}{dt^2} = 0$$

$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{i}{C} + 40 \sin 4t = 0$$

2nd order differential equation:

$$i(t) = A_i \cos(4t + \phi_i) \quad V_C(t) = A_C \cos(4t + \phi_C)$$

$$V_R(t) = A_R \cos(4t + \phi_R) \quad V_L(t) = A_L \cos(4t + \phi_L)$$



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AC circuit analysis with Phasors?

Method #2: Frequency or **Phasor** domain
Transform: time based functions to frequency
(phasors/complex)

$$i(t) \Rightarrow I \quad R \Rightarrow Z_R$$

$$C \Rightarrow Z_C$$

$$V_s \Rightarrow V_s \quad L \Rightarrow Z_L$$

KVL:

$$-\overline{V}_s + \overline{V}_R + \overline{V}_C + \overline{V}_L = 0$$

Ohms Law

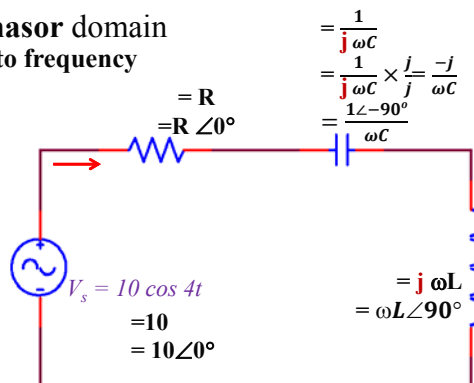
$$-\overline{V}_s + IZ_R + IZ_C + IZ_L = 0$$

$$-\overline{V}_s + I(Z_R + Z_C + Z_L) = 0$$

$$-10 \angle 0^\circ + I \left(R + \frac{1}{j\omega C} + j\omega L \right) = 0$$

$$I = \frac{10 \angle 0^\circ}{R \angle 0^\circ + \frac{1}{\omega C} \angle -90^\circ + \omega L \angle 90^\circ}$$

$$I = A_i \angle \phi_i^\circ$$



Where

A_i – amplitude of the signal

$\angle \phi_i^\circ$ – phase of the signal

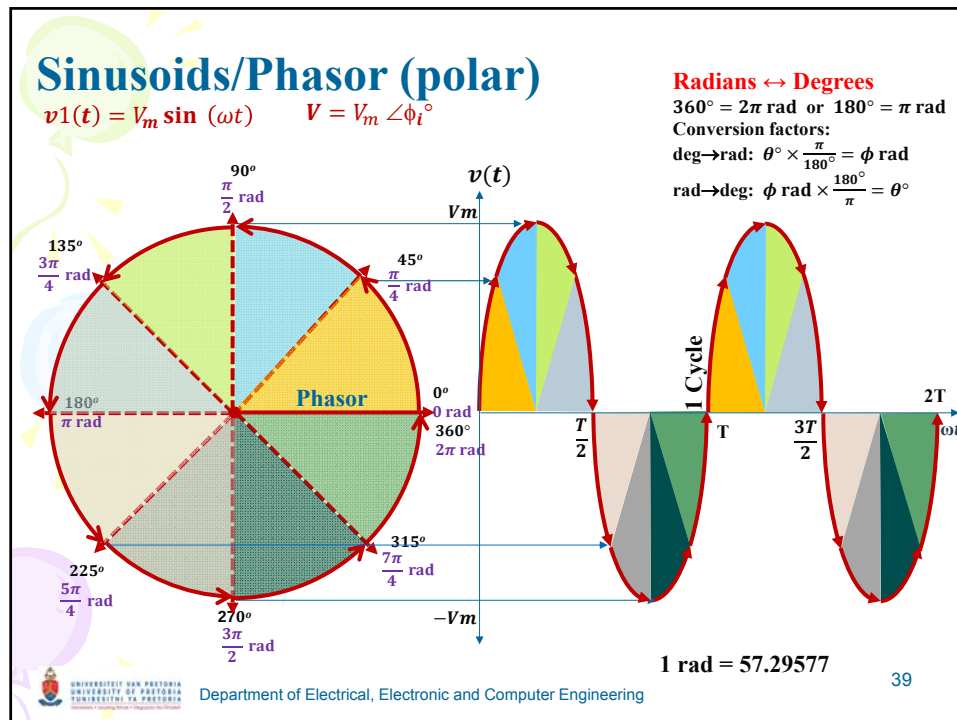
$$A_i \angle \phi_i^\circ \rightarrow i(t) = A_i \cos(\omega t + \phi_i)$$



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9.3 Phasors

- A phasor is a complex number that represents the amplitude and phase of a sinusoid (COSINE).

Find the rectangular and Exponential form of

$$z = 2.5 \angle -30^\circ$$

$$r = 2.5 \text{ and } \phi = -30^\circ$$

$$x = r \cos \phi$$

$$= 2.5 \cos 30^\circ = 2.17$$

$$y = r \sin \phi$$

$$= 2.5 \sin 30^\circ = 1.25$$

Rectangular form

$$z = 2.17 + j 1.25$$

Exponential Form

$$z = 2.5 e^{j -30^\circ}$$

$$x = r \cos \phi \quad y = r \sin \phi$$

$$r = \sqrt{x^2 + y^2} \quad \phi = \tan^{-1} \left(\frac{y}{x} \right)$$

Phasors: On the complex plane

**Example 1: Consider $z = 1$,
Write its three forms**

Rectangular:

$$z = 1 + j0$$

Polar Form

$$z = 1 \angle 0^\circ$$

Exponential Form

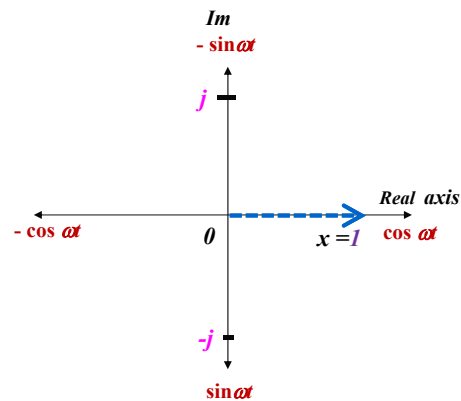
$$z = 1 e^{j0^\circ}$$

Plot $z = 1$

$$z = \cos(\alpha + 0)$$

$$x = r \cos \phi \quad y = r \sin \phi$$

$$r = \sqrt{x^2 + y^2} \quad \phi = \tan^{-1} \left(\frac{y}{x} \right)$$



**Example 2: Consider $z = -j$,
Write its three forms and plot.**

Rectangular:

$$z = 0 - j$$

Polar Form

$$r = \sqrt{0^2 + 1^2} = \pm 1$$

$$\phi = \tan^{-1} \left(\frac{1}{0} \right) = -90^\circ$$

$$z = 1 \angle -90^\circ$$

Exponential Form

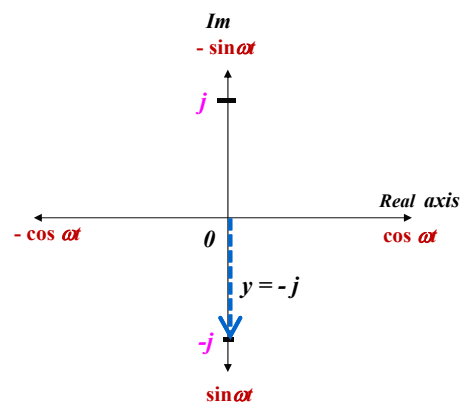
$$z = 1 e^{j-90^\circ}$$

Plot $z = -j$

$$z = \cos(\alpha - 90^\circ)$$

$$x = r \cos \phi \quad y = r \sin \phi$$

$$r = \sqrt{x^2 + y^2} \quad \phi = \tan^{-1} \left(\frac{y}{x} \right)$$



**Example 3: Consider $z = -5 \cos (\omega t - 135^\circ)$,
Write its three forms and plot.**

$$z = 5 \cos (\omega t - 135^\circ + 180^\circ) \\ = 5 \cos (\omega t + 45^\circ)$$

Polar Form

$$z = 5 \angle 45^\circ$$

Exponential Form

$$z = 5 e^{j 45^\circ}$$

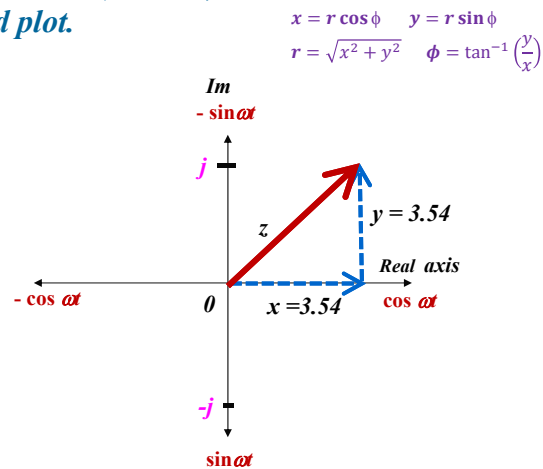
Rectangular:

$$z = x + j y$$

$$x = r \cos \phi = 5 \cos 45^\circ \\ = 3.54$$

$$y = r \sin \phi = 5 \sin 45^\circ \\ = 3.54$$

$$z = 3.54 + j 3.54$$



Eg: Consider the sinusoid: $-5 \cos (\omega t - 135^\circ)$. Write its three complex forms

Sinusoid $\Rightarrow$ Phasor: Method #1 Analytical

First change z to

$$+A_m \cos (\omega t + \phi)$$

$$z = -5 \cos (\omega t - 135^\circ) \\ = 5 \cos (\omega t - 135^\circ + 180^\circ)$$

$$z = 5 \cos (\omega t + 45^\circ)$$

Rectangular: $z = x + j y$

$$x = r \cos \phi \quad y = r \sin \phi$$

$$x = 5 \cos 45^\circ = 3.54$$

$$y = 5 \sin 45^\circ = 3.54$$

$$\therefore z = 3.54 + j 3.54$$

Polar Form: $z = r \angle \phi$

$$z = 5 \cos (\omega t + 45^\circ)$$

$$r = 5$$

$$\phi = 45^\circ$$

$$z = 5 \angle 45^\circ$$

Exponential Form: $z = r e^{j \phi}$

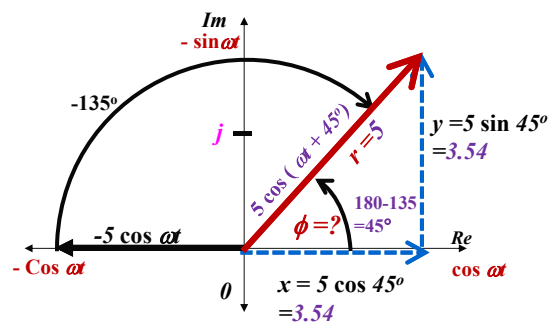
$$z = 5 \cos (\omega t + 45^\circ)$$

$$z = 5 e^{j 45^\circ}$$

$$\begin{aligned} +A_m \sin (\omega t \pm \phi) &\Rightarrow +A_m \cos (\omega t \pm \phi - 90^\circ) \\ -A_m \sin (\omega t \pm \phi) &\Rightarrow +A_m \cos (\omega t \pm \phi + 90^\circ) \\ -A_m \cos (\omega t \pm \phi) &\Rightarrow +A_m \cos (\omega t \pm \phi \pm 180^\circ) \end{aligned}$$

Method #2 Using complex plane

$$z = -5 \cos (\omega t - 135^\circ)$$

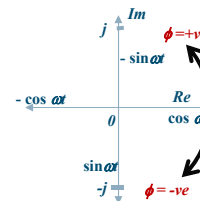


Homework:

For the sinusoids given below write their three complex forms.

1. $-5 \cos (\omega t + 235^\circ)$
2. $5 \cos (\omega t + 235^\circ)$
3. $-5 \cos (\omega t - 235^\circ)$
4. $5 \cos (\omega t - 235^\circ)$

5. $-5 \sin (\omega t - 35^\circ)$
6. $5 \sin (\omega t - 35^\circ)$
7. $-5 \sin (\omega t + 35^\circ)$
8. $5 \sin (\omega t + 35^\circ)$



$$\begin{aligned}
 +A_m \sin (\omega t \pm \phi) &\Rightarrow +A_m \cos (\omega t \pm \phi - 90^\circ) \\
 -A_m \sin (\omega t \pm \phi) &\Rightarrow +A_m \cos (\omega t \pm \phi + 90^\circ) \\
 -A_m \cos (\omega t \pm \phi) &\Rightarrow +A_m \cos (\omega t \pm \phi \pm 180^\circ)
 \end{aligned}$$

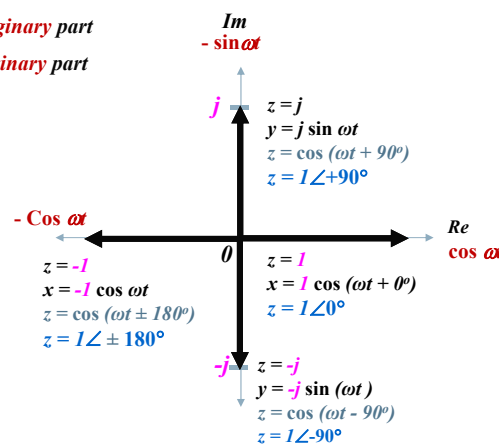
Special sinusoids/phasors

$z = 1$ z has only +ve real part

$z = -1$ z has only -ve real part

$z = j$ z has only +ve imaginary part

$z = -j$ z has only -ve imaginary part



Phasor Conversion

$$j = \sqrt{-1}$$

$$= -\sin \omega t$$

Polar form: $= 1 \angle 90^\circ$

$$\frac{1}{j} = ?$$

$$= \frac{1}{\sqrt{-1}}$$

$$= \frac{1}{\sqrt{-1}} \times \frac{\sqrt{-1}}{\sqrt{-1}}$$

$$= \frac{\sqrt{-1}}{\sqrt{-1}}$$

$$= \frac{j}{-1}$$

$$\frac{1}{j} = -j$$

$$= \sin \omega t$$

$$= \cos(\omega t - 90^\circ)$$

Polar form: $= 1 \angle -90^\circ$

$$j + j = ?$$

$$= 2j = 2\sqrt{-1}$$

$$2j^2 = ?$$

$$= 2(\sqrt{-1})^2 = -2$$

$$= -2 \cos \omega t$$

$$= 2 \cos(\omega t \pm 180^\circ)$$

Polar form: $= 2 \angle \pm 180^\circ$

$$z = 1 + j$$

$$x = 1, y = 1$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) = 45^\circ$$

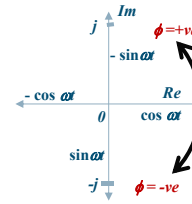
$$r = \sqrt{1^2 + 1^2} = \pm\sqrt{2}$$

$$\therefore z = \sqrt{2} \angle 45^\circ$$

$$z = 1 - j$$

Similarly

$$z = \sqrt{2} \angle -45^\circ$$



AC signal as a Phasor

Euler's identity:

$$e^{\pm j\phi} = \cos \phi \pm j \sin \phi$$

$$\operatorname{Re}\{e^{j\phi}\} = \cos \phi$$

$$\operatorname{Im}\{e^{j\phi}\} = \sin \phi$$

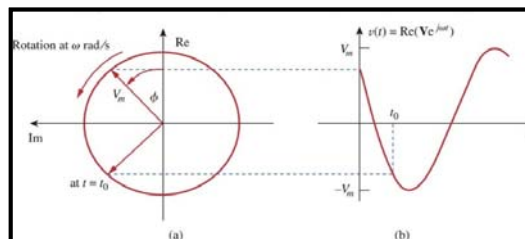
$$\text{Let } v(t) = V_m \cos(\omega t + \phi)$$

$$= \operatorname{Re}\{V_m e^{j(\omega t + \phi)}\}$$

$$= \operatorname{Re}\{V_m e^{j\phi} e^{j\omega t}\}$$

$$\text{Let } \mathbf{V} = V_m e^{j\phi}$$

$$v(t) = \operatorname{Re}\{\mathbf{V} e^{j\omega t}\}$$



$$\mathbf{V} = \bar{\mathbf{V}} = V_m e^{j\phi} = V_m \angle \phi$$

$$= V_m \cos \phi + j V_m \sin \phi$$

$$= x + jy$$

Term $e^{j\omega t} = \cos \omega t + j \sin \omega t$

Is Implicitly present

PP 9.4 Express these sinusoids as phasors:

a) $v = 7 \cos(2t + 40^\circ) \text{ V}$

Since v is in $+\cos(\omega t)$, no transformation is required

$$\mathbf{V} = 7 \angle 40^\circ \text{ V}$$

b) $i = -4 \sin(10t + 10^\circ) \text{ A}$

Change i to $+\cos$

$$= 4 \cos(10t + 10^\circ + 90^\circ)$$

$$= 4 \cos(10t + 100^\circ)$$

$$\mathbf{I} = 4 \angle 100^\circ \text{ A}$$



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PP Find the sinusoids corresponding to these phasors:

a) $\mathbf{V} = -10 \angle 30^\circ \text{ V}$

$$v = -10 \cos(\omega t + 30^\circ) \text{ V}$$

$$v = 10 \cos(\omega t + 30 - 180^\circ) \text{ V}$$

$$= 10 \cos(\omega t - 150^\circ) \text{ V}$$

Or

$$v = 10 \cos(\omega t + 30 + 180^\circ) \text{ V}$$

$$= 10 \cos(\omega t + 210^\circ) \text{ V}$$

ϕ is outside $\pm 180^\circ$ range.

b) $\mathbf{I} = j(5 - j12) \text{ A}$

$$\mathbf{I} = j5 + 12$$

$$= \sqrt{12^2 + 5^2} \angle \tan^{-1}\left(\frac{5}{12}\right)$$

$$= 13 \angle 22.62^\circ$$

$$i = 13 \cos(\omega t + 22.62^\circ) \text{ A}$$



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PP Find the voltage $v(t)$ in a circuit described by the integrodifferential equation:

$$2 \frac{dv}{dt} + 5v + 10 \int v dt = 50 \cos(5t - 30^\circ)$$

$$\int v dt \Leftrightarrow \frac{V}{j\omega}$$

$$\frac{dv}{dt} \Leftrightarrow j\omega V$$

$$2j\omega V + 5V + 10 \frac{V}{j\omega} = 50 \angle -30^\circ$$

$$\omega = 5 \text{ rad/sec}$$

$$j10V + 5V + 10 \frac{V}{j5} = 50 \angle -30^\circ \quad \} \times j5$$

$$j5 \times j10V + j5 \times 5V + 10V = j5 \times 50 \angle -30^\circ$$

$$j^2 50V + j25V + 10V = 5 \angle 90^\circ \times 50 \angle -30^\circ$$

$$-50V + j25V + 10V = 250 \angle 60^\circ$$

$$-40V + j25V = 250 \angle 60^\circ$$

$$V = \frac{250 \angle -120^\circ}{-40 + j25} = \frac{250 \angle 60^\circ}{47.17 \angle 148^\circ} = 5.3 \angle 268^\circ$$

$$V = 5.3 \angle -88^\circ$$

$$v(t) = 5.3 \cos(5t - 88^\circ) \text{ A}$$



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L#4 9.5 Impedance and Admittance

- The impedance Z of a circuit is the ratio of the phasor voltage V to the phasor current I , measured in ohms Ω .

$$Z = \frac{V}{I} = R + jX$$

where

$R = \text{Re of } Z = \text{the resistance and}$

$X = \text{Im of } Z = \text{the reactance.}$

Positive X is for L and negative X is for C .

- The admittance Y is the reciprocal of impedance, measured in siemens (S).

$$Y = \frac{1}{Z} = \frac{I}{V}$$

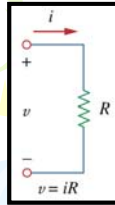


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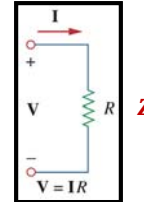
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Phasors/Circuit Elements and Impedance (Z)

Resistor:



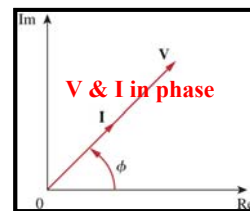
$$\begin{aligned}
 i(t) &= I_m \cos(\omega t + \phi) \\
 I &= I_m \angle \phi \\
 v(t) &= R i(t) \\
 &= R I_m \cos(\omega t + \phi) \\
 V &= R I_m \angle \phi \\
 V &= R I \\
 \frac{V}{I} &= R = Z
 \end{aligned}$$



Compare the phase of current and voltage

$$\begin{aligned}
 I &= I_m \angle \phi \\
 V &= R I_m \angle \phi
 \end{aligned}$$

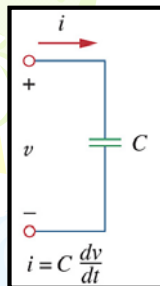
There is no change in the phase of the current and voltage
Current and voltage are **in-phase** (same phase)



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Capacitor phase and its Impedance (Z_C):



$$\begin{aligned}
 v(t) &= V_m \cos(\omega t + \phi); \quad V = V_m \angle \phi \\
 i(t) &= C \frac{dv}{dt} = C \frac{d}{dt} V_m \cos(\omega t + \phi) \\
 &= -V_m C \omega \sin(\omega t + \phi) \\
 &= -\omega C V_m \sin(\omega t + \phi) \\
 &= \omega C V_m \cos(\omega t + \phi + 90^\circ) \\
 I &= \omega C V_m \angle (\phi + 90^\circ) \\
 &= \omega C V_m \angle (\phi) 1 \angle (90^\circ) \\
 &= \omega C V 1 \angle (90^\circ)
 \end{aligned}$$

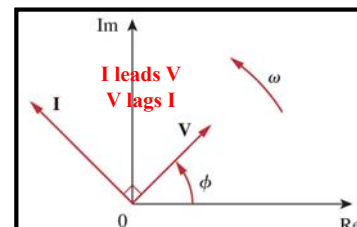
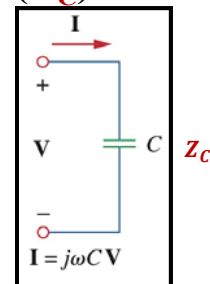
$$\begin{aligned}
 I &= j\omega C V \\
 \frac{V}{I} &= \frac{1}{j\omega C} = Z_C
 \end{aligned}$$

Compare the current and voltage phase

$$V = V_m \angle \phi$$

$$I = \omega C V_m \angle (\phi + 90^\circ)$$

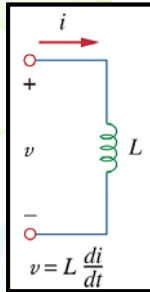
It is clear that current leads voltage by 90°



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Inductor phase and its Impedance (Z_L):



$$i(t) = I_m \cos(\omega t + \phi); \quad I = I_m \angle \phi$$

$$v(t) = L \frac{di}{dt} = L \frac{d}{dt} I_m \cos(\omega t + \phi)$$

$$= -\omega L I_m \sin(\omega t + \phi)$$

$$= \omega L I_m \cos(\omega t + \phi + 90^\circ)$$

$$V = \omega L I_m \angle(\phi + 90^\circ)$$

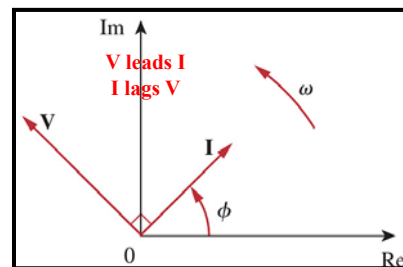
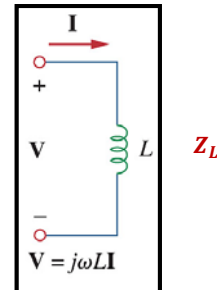
$$= \omega L I_m \angle(\phi) 1 \angle(90^\circ)$$

$$= \omega L I 1 \angle(90^\circ)$$

$$= \omega L I j$$

$$V = j\omega L I$$

$$\frac{V}{I} = j\omega L = Z_L$$



Compare the current and voltage phase

$$I = I_m \angle \phi$$

$$V = \omega L I_m \angle(\phi + 90^\circ)$$

It is clear that voltage leads current by 90°

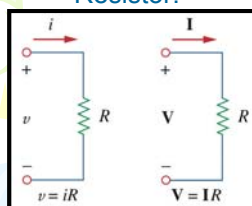


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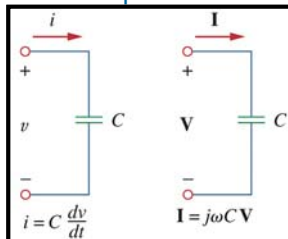
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Phasors and Circuit Elements- Summary

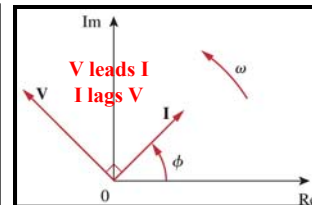
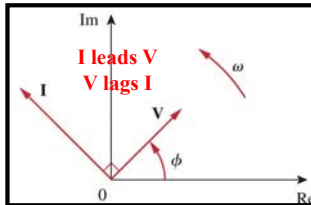
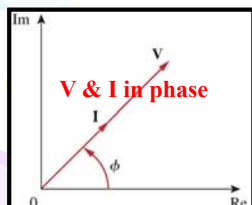
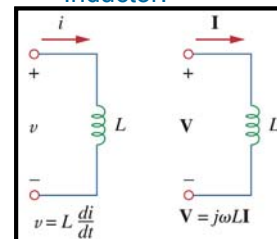
Resistor:



Capacitor:



Inductor:



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Voltage-Current Relationship

Summary of voltage-current relationship

Elem ent	Time domain	Frequency domain	Impedance	Admittance
R	$v(t) = R i(t)$	$V = R I$	$Z = R$	$Y = \frac{1}{R}$
L	$v(t) = L \frac{di}{dt}$	$V = j\omega L I$	$Z_L = j\omega L$	$Y_L = \frac{1}{j\omega L}$
C	$i(t) = C \frac{dv}{dt}$	$V = \frac{I}{j\omega C}$	$Z_C = \frac{1}{j\omega C}$	$Y_C = j\omega C$



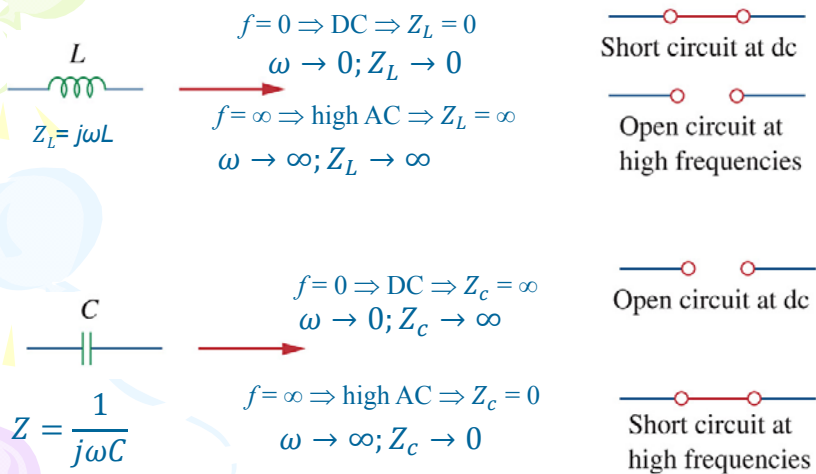
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9.5 Impedance (**Z**) and Admittance (**Y**) Behaviour

Remember: $\omega = 2\pi f$



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Impedance (Z) and Admittance (Y)

$$Z = R + jX = |Z| \angle \theta$$

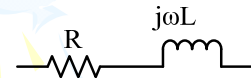
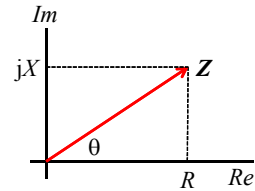
X - Reactance

$$|Z| = \sqrt{R^2 + X^2}$$

$$\theta = \tan^{-1} \frac{X}{R}$$

$$Y = \frac{1}{Z} = G + jB$$

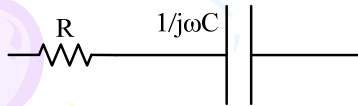
G - Conductance
B - Susceptance



$$Z = R + j\omega L$$

$$= R + jX_L$$

$$\therefore X_L = \omega L$$



$$Z = R + \frac{1}{j\omega C}$$

$$= R - j \frac{1}{\omega C}$$

$$\therefore X_C = -\frac{1}{\omega C}$$



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9.5 Impedance and Admittance

After we know how to convert RLC components from time to phasor domain, we can transform a time domain circuit into a phasor/frequency domain circuit.

Hence, we can apply the KCL laws and other theorems to directly set up phasor equations involving our target variable(s) for solving.



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PP 9.8

If voltage $v = 10 \cos(100t + 30^\circ)$ V is applied to a $50 \mu\text{F}$ capacitor, calculate the current through the capacitor.

Step 1: Convert from time to phasor domain

$$v = 10 \cos(100t + 30^\circ) \Rightarrow \mathbf{V} = 10 \angle 30^\circ \text{ V}$$

$$\begin{aligned} \mathbf{Z} &= jX_C = j(-1/\omega C) = -j/(100 \times 50 \times 10^{-6}) \\ &= -j200 \Omega \\ &= 200 \angle -90^\circ \Omega \end{aligned}$$

Step 2: Apply usual Laws and Theorems

$$\begin{aligned} \mathbf{I} &= \mathbf{V} / \mathbf{Z} = 10 \angle 30^\circ / 200 \angle -90^\circ \\ &= 50 \angle (30 - (-90))^\circ \text{ mA} \\ &= 50 \angle 120^\circ \text{ mA} \end{aligned}$$

Step 3: Convert from phasor to time domain

$$i(t) = 50 \cos(100t + 120^\circ) \text{ mA}$$



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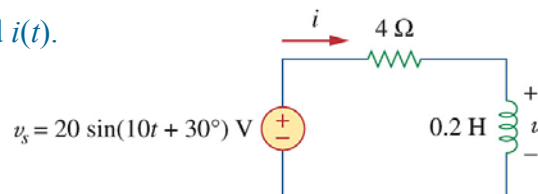
PP 9.9 Determine $v(t)$ and $i(t)$.**Step 1:**

Convert from time to phasor domain

$$\begin{aligned} v_s &= 20 \sin(10t + 30^\circ) \\ &= 20 \cos(10t - 60^\circ) \end{aligned}$$

$$\Rightarrow \mathbf{V}_s = 20 \angle -60^\circ \text{ V}$$

$$\begin{aligned} \mathbf{Z} &= R + jX_L \\ &= R + j\omega L \\ &= 4 + j(10 \times 0.2) \\ &= 4 + j2 \Omega \\ &= 4.47 \angle 26.57^\circ \Omega \end{aligned}$$

**Step 3: Convert from phasor to time domain**

$$\mathbf{I} = 4.47 \angle -86.57^\circ \text{ A}$$

$$i(t) = 4.47 \cos(10t - 86.57^\circ) \text{ A}$$

$$\mathbf{V}_L = 8.944 \angle 3.43^\circ \text{ V}$$

$$v_L(t) = 8.944 \cos(10t + 3.43^\circ) \text{ V}$$

Step 2: Apply usual Laws and Theorems

$$\begin{aligned} \mathbf{I} &= \mathbf{V} / \mathbf{Z} = 20 \angle -60^\circ / 4.47 \angle 26.57^\circ \\ &= 4.47 \angle -86.57^\circ \text{ A} \end{aligned}$$

$$\mathbf{V}_L = \mathbf{I} \mathbf{Z}_L = 4.47 \angle -86.57^\circ \times 2 \angle 90^\circ$$

$$\mathbf{V}_L = 8.944 \angle 3.43^\circ \text{ V}$$



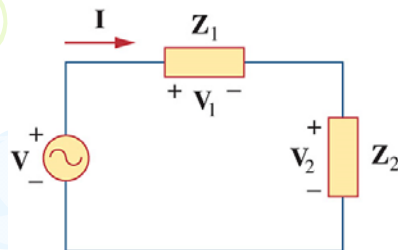
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L#5 AC Circuit analysis

Laws and Theorems

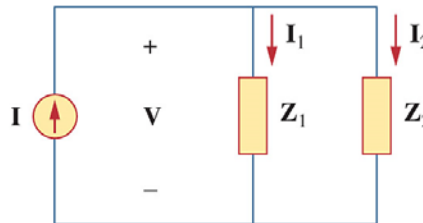


KVL: $-V + V_1 + V_2 = 0$

$V = IZ_1 + IZ_2$

Z in series: $Z_T = Z_1 + Z_2$

Voltage division: $V_1 = V \frac{Z_1}{Z_1 + Z_2}$



KCL: $I = I_1 + I_2$

Z in //: $Z_{||} = \frac{Z_1 Z_2}{Z_1 + Z_2}$

Current division: $I_1 = I \frac{Z_2}{Z_1 + Z_2}$



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2.7 Wye-Delta Transformations

Y inside Δ

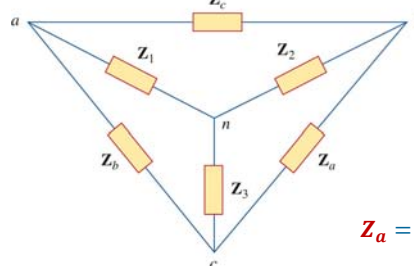
Delta → Wye

Δ → Y

$$Z_1 = \frac{Z_b Z_c}{Z_a + Z_b + Z_c}$$

$$Z_2 = \frac{Z_a Z_c}{Z_a + Z_b + Z_c}$$

$$Z_3 = \frac{Z_a Z_b}{Z_a + Z_b + Z_c}$$



Wye → Delta

Y → Δ

$$Z_a = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_1}$$

$$Z_b = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_2}$$

$$Z_c = \frac{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1}{Z_3}$$

For a BALANCED load:

$$Z_\Delta = 3Z_Y$$

Or

$$Z_Y = \frac{1}{3}Z_\Delta$$



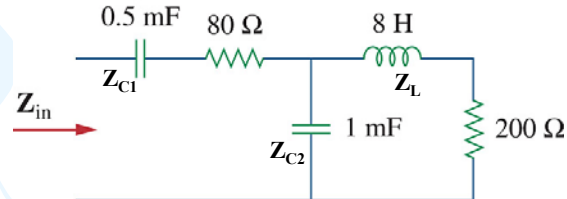
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PP 9.10Determine the input impedance of the circuit at $\omega = 10 \text{ rad/s}$

$$\begin{aligned}
 Z_{C1} &= jX_{C1} \\
 &= j(-1/\omega C1) \\
 &= -j/(10 \times 0.5 \times 10^{-3}) \\
 &= -j200 \Omega \\
 &= \mathbf{200 \angle -90^\circ \Omega}
 \end{aligned}$$

$$\begin{aligned}
 Z_L &= jX_L = j\omega L \\
 &= j(10 \times 8) \\
 &= j80 \Omega \\
 &= \mathbf{80 \angle 90^\circ \Omega}
 \end{aligned}$$

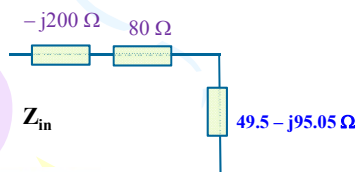
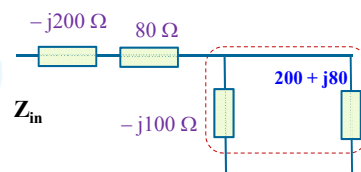
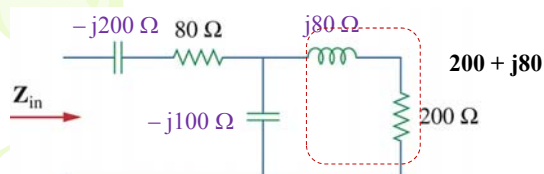


$$\begin{aligned}
 Z_{C2} &= jX_{C2} \\
 &= j(-1/\omega C2) \\
 &= -j/(10 \times 1 \times 10^{-3}) \\
 &= -j100 \Omega \\
 &= \mathbf{100 \angle -90^\circ \Omega}
 \end{aligned}$$



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PP 9.10 ...

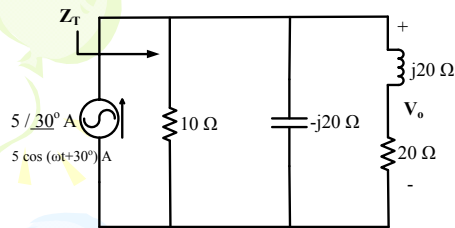
$$\begin{aligned}
 &(200 + j80) \parallel -j100 \\
 &= \frac{100 \angle -90^\circ \times 215.41 \angle 21.8^\circ}{-j100 + (200 + j80)} \\
 &= \frac{21.54 \times 10^3 \angle -68.2^\circ}{201 \angle -5.71^\circ} \\
 &= 107.17 \angle -62.5^\circ \\
 &= \mathbf{49.5 - j95.05 \Omega}
 \end{aligned}$$

$$\begin{aligned}
 Z_{in} &= -j200 + 80 + 49.5 - j95.05 \\
 &= 129.51 - j295.05 \\
 &= \mathbf{322.22 \angle -66.30^\circ \Omega} \\
 &= \mathbf{129.5 - j295 \Omega}
 \end{aligned}$$



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EP – Calculate Z_T and V_o .

$$V_o = I \times Z_T$$

$$= 5\angle 30^\circ \times 7.84\angle -11.34^\circ$$

$$= 39.2\angle 18.66^\circ \text{ V}$$

$$= 37.14 + j12.5 \text{ V}$$

$$Z_T = 10 \parallel -j20 \parallel (20 + j20)$$

$$= \frac{-j200}{10 - j20} \parallel (20 + j20)$$

$$= 8.94\angle -26.6^\circ \parallel 28.28\angle 45^\circ$$

$$= \frac{8.94\angle -26.6^\circ \times 28.28\angle 45^\circ}{8.94\angle -26.6^\circ + 28.28\angle 45^\circ}$$

$$= \frac{252.86\angle 18.4^\circ}{32.24\angle 29.74^\circ}$$

$$= 7.84\angle -11.34^\circ$$

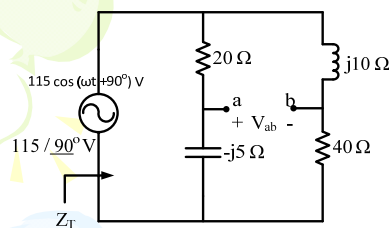
$$= 7.7 - j1.54 \Omega$$



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EP – Calculate Z_T and V_{ab} . Give final answer in time domain.

$$Z_T = (20 - j5) \parallel (40 + j10)$$

$$= \frac{(20 - j5) \times (40 + j10)}{20 - j5 + 40 + j10}$$

$$= \frac{850}{60 + j5}$$

$$= 14.1\angle -4.76^\circ$$

$$Z_T = 14.1 - j1.17 \Omega$$

$$V_{ab} = V_a - V_b$$

$$= V_{-j5} - V_{40}$$

$$V_{-j5} = \frac{-j5}{20 - j5} \times 115\angle 90^\circ \text{ - voltage divider}$$

$$= \frac{550}{20 - j5}$$

$$= 26.7\angle 14.04^\circ \text{ V}$$

$$= 25.9 + j6.5 \text{ V}$$

$$V_{40} = \frac{40}{40 + j10} \times 115\angle 90^\circ \text{ - voltage divider}$$

$$= \frac{4600\angle 90^\circ}{40 + j10}$$

$$= 111.57\angle 76^\circ \text{ V}$$

$$= 27.06 + j108.24 \text{ V}$$

$$V_{ab} = (25.9 + j6.5) - (27.06 + j108.24)$$

$$= -1.16 - j101.7 \text{ V}$$

$$= 101.7\angle -90.65^\circ$$

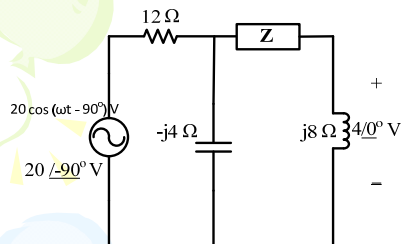
$$v_{ab}(t) = 101.7 \cos(\omega t - 90.65^\circ) \text{ V}$$



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EP – Calculate Z.

$$\begin{aligned}
 Z_T &= ((Z+j8)||(-j4))+12 \\
 &= \frac{(Z+j8)(-j4)}{Z+j8-j4} + 12 \\
 &= \frac{-j4Z+32}{Z+j4} + 12 \\
 &= \frac{-j4Z+32+12(Z+j4)}{Z+j4} \\
 &= \frac{-j4Z+32+12Z+j48}{Z+j4}
 \end{aligned}$$

$$I_Z = \frac{4}{j8} = \frac{1}{j2} = -j0.5$$

$$I_Z = \frac{4}{Z+j8-j4} \times I \quad \text{- current divider}$$

$$= \frac{4}{Z+j4} \times I = \frac{1}{j2}$$

$$I = -j0.5 \times \frac{Z+j4}{4} \quad (1)$$

$$\text{Also } I = \frac{V}{Z_T} = \frac{20 \angle -90}{Z_T}$$

$$= \frac{-j20}{\frac{-j4Z+32+12Z+j48}{Z+j4}} \quad (2)$$

$$\frac{-j20}{\frac{-j4Z+32+12Z+j48}{Z+j4}} = -j0.5 \times \frac{Z+j4}{4}$$

$$-j20 = -j0.5 \times \frac{Z+j4}{4} \times \frac{-j4Z+32+12Z+j48}{Z+j4}$$

$$Z = 3.55 - j16.15 \Omega$$